

N-CHANNEL SILICON FIELD-EFFECT TRANSISTORS

Symmetrical N-channel planar epitaxial junction field-effect transistors in a plastic TO-92 variant; intended for hi-fi amplifiers and other audio-frequency equipment.

QUICK REFERENCE DATA

Drain-source voltage	$\pm V_{DS}$	max.	30 V
Total power dissipation up to $T_{amb} = 75\text{ }^{\circ}\text{C}$	P_{tot}	max.	300 mW
Junction temperature	T_j	max.	150 $^{\circ}\text{C}$
Drain current $V_{DS} = 15\text{ V}; V_{GS} = 0$	I_{DSS}		2 to 12 mA
Transfer admittance (common source) $V_{DS} = 15\text{ V}; V_{GS} = 0; f = 1\text{ kHz}$	$ Y_{fs} $	typ.	3,5 mS
Noise figure at $V_{DS} = 15\text{ V}; V_{GS} = 0$ $f = 1\text{ kHz}; R_G = 1\text{ M}\Omega$	F	<	2 dB

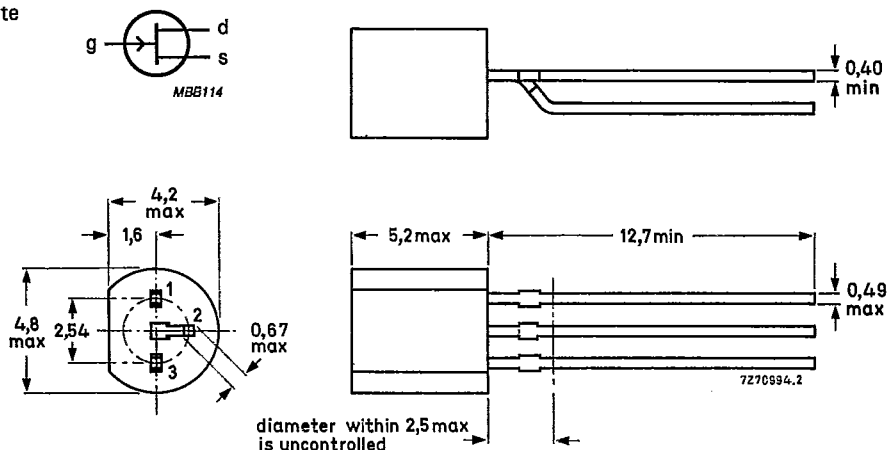
MECHANICAL DATA

Dimensions in mm

Fig. 1 TO-92 variant.

Pinning:

- 1 = drain
- 2 = source
- 3 = gate



Note: Drain and source are interchangeable

RATINGS Limiting values in accordance with the Absolute Maximum System (IEC134)

Drain-source voltage	$\pm V_{DS}$	max.	30	V
Drain-gate voltage (open source)	V_{DGO}	max.	30	V
Gate-source voltage (open drain)	$-V_{GSO}$	max.	30	V
Gate current	I_G	max.	10	mA
Total power dissipation up to $T_{amb} = 75^\circ\text{C}$	P_{tot}	max.	300	mW
Storage temperature range	T_{stg}	-65 to +150		$^\circ\text{C}$
Junction temperature	T_j	max.	150	$^\circ\text{C}$

THERMAL RESISTANCE

From junction to ambient in free air	$R_{th\ j-a}$	=	250	K/W
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CHARACTERISTICS

 $T_j = 25\text{ }^{\circ}\text{C}$ unless otherwise specified

Gate cut-off current

$-V_{GS} = 20\text{ V}; V_{DS} = 0$

$-I_{GSS}$

BC264A	B	C	D
< 5	5	5	5 nA

Drain current

$V_{DS} = 15\text{ V}; V_{GS} = 0$

I_{DSS}

> 2,0	3,5	5,0	7,0 mA
< 4,5	6,5	8,0	12,0 mA

Gate-source breakdown voltage

$-I_G = 1\text{ }\mu\text{A}; V_{DS} = 0$

$-V_{(BR)GSS}$

> 30	30	30	30 V
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Gate-source voltage

$I_D = 200\text{ }\mu\text{A}; V_{DS} = 15\text{ V}$

$-V_{GS}$

> 0,4	0,4	0,4	0,4 V
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$I_D = 1,0\text{ mA}; V_{DS} = 15\text{ V}$

$-V_{GS}$

> 0,2	—	—	— V
< 1,2	—	—	— V

$I_D = 1,5\text{ mA}; V_{DS} = 15\text{ V}$

$-V_{GS}$

> —	0,4	—	— V
< —	1,4	—	— V

$I_D = 2,5\text{ mA}; V_{DS} = 15\text{ V}$

$-V_{GS}$

> —	—	0,5	— V
< —	—	1,5	— V

$I_D = 3,5\text{ mA}; V_{DS} = 15\text{ V}$

$-V_{GS}$

> —	—	—	0,6 V
< —	—	—	1,6 V

Gate-source cut-off voltage

$I_D = 10\text{ nA}; V_{DS} = 15\text{ V}$

$-V_{(P)GS}$

> 0,5	0,5	0,5	0,5 V
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y-parameters at $T_{amb} = 25\text{ }^{\circ}\text{C}$

$V_{DS} = 15\text{ V}; V_{GS} = 0; f = 1\text{ kHz}$

Transfer admittance

$|y_{fs}|$

> 2,5	3,0	3,5	4,0 mS
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$V_{DS} = 15\text{ V}; -V_{GS} = 1\text{ V}; f = 1\text{ MHz}$

Input capacitance

C_{is}

typ.	4,0	pF
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Feedback capacitance

C_{rs}

typ.	1,2	pF
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Output capacitance

C_{os}

typ.	1,6	pF
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Noise figure at $f = 1\text{ kHz}; R_G = 1\text{ M}\Omega$

$V_{DS} = 15\text{ V}; V_{GS} = 0; T_{amb} = 25\text{ }^{\circ}\text{C}$

F

typ.	0,5	dB
<	2	dB

Equivalent noise voltage at $T_{amb} = 25\text{ }^{\circ}\text{C}$

$V_{DS} = 15\text{ V}; V_{GS} = 0; f = 10\text{ Hz}$

$V_n/\sqrt{B}$

typ.	40	nV/ $\sqrt{\text{Hz}}$
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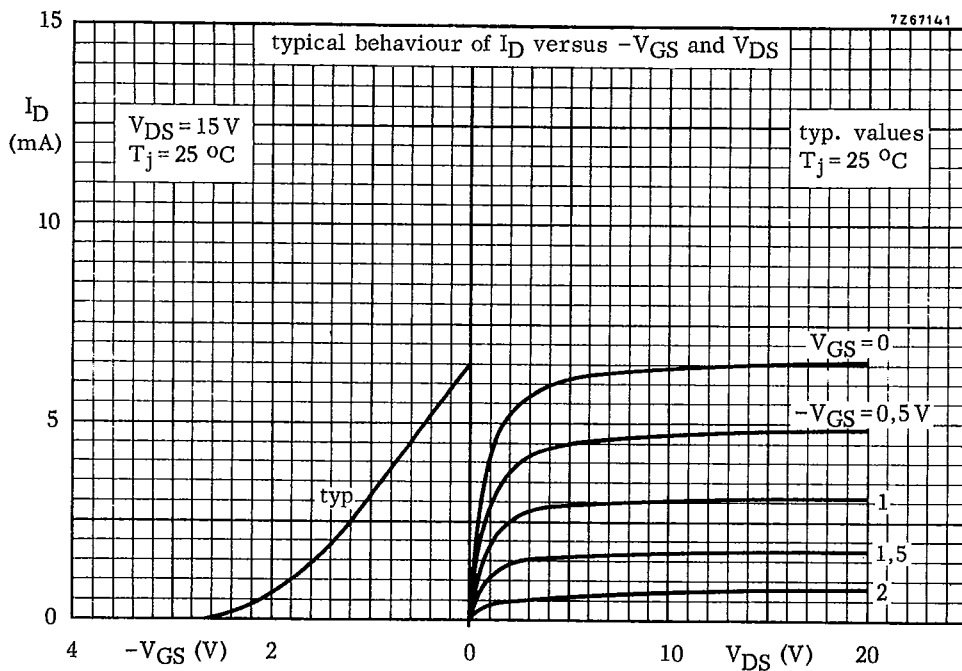


Fig. 2

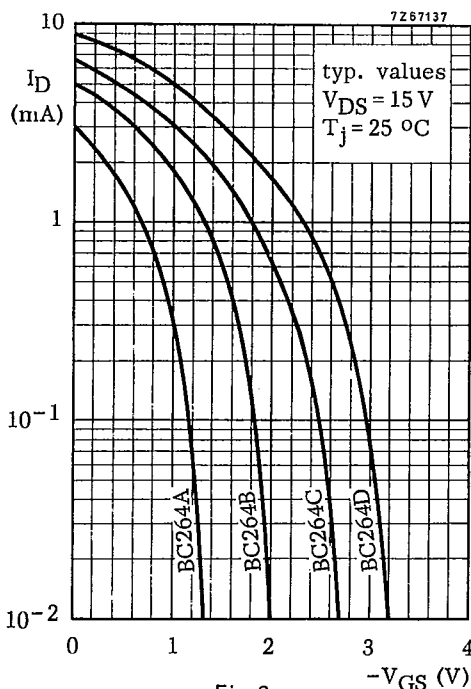


Fig. 3

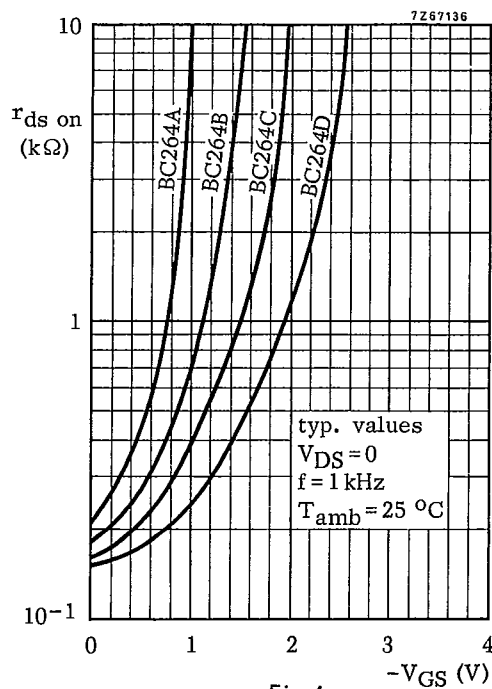


Fig. 4

